

Little Red Dots as cosmic-ray electron sources: literature status, competing hypotheses, and a recipe for estimating the index and normalization of their electron spectrum

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companion to `manuscript/manuscript.tex` (cosmic-ray electrons in the IGM, $z = 5.5\text{--}20$)

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Abstract

This report answers four questions in order. **(1)** It surveys the Little Red Dot (LRD) literature to the end of August 2026. **(2)** It lists the nine hypotheses currently on the table for what LRDs are. **(3)** It identifies which of those objects can accelerate cosmic rays and gives an explicit, closed recipe for estimating the index s and the normalization K of a power-law cosmic-ray electron spectrum $N(\gamma) = K\gamma^{-s}$, along both a forward (energetic) and an inverse (radio-limit) route, with a worked fiducial. **(4)** Every algebraic step is verified symbolically with SYMPY (`verify_algebra.py`); every number is produced by `lrd_cr_numbers.py`, which also reproduces Table 3 of the companion manuscript to 0.14%. The central finding is that the radio non-detections of LRDs do *not* constrain their cosmic-ray electron content, because inverse-Compton losses in the dense cocoon suppress synchrotron emission by a factor $1 + U_{\text{rad}}/U_B \sim 10^2\text{--}10^5$; the constraint they do provide is three orders of magnitude weaker than the fiducial prediction. Folding the escaping electrons through the cascade yield table of the companion manuscript gives $2 \times 10^{-6}\text{--}10^{-4}$ ionizations per hydrogen atom and $\Delta T \sim 0.1\text{--}18$ K for plausible parameters.

1 Scope and rules

This report follows the conventions of the companion manuscript: every physical statement is attributed to a real publication, cited in BIBTEX form, with a free-access link collected in Sect. 9; figures are produced by `.py` files in this directory; and no astrophysical hypothesis is adopted without it being flagged as such. Assumptions that are mine rather than the literature’s are marked [A] throughout.

2 The recent literature on Little Red Dots

LRDs were defined observationally as compact, red, high-redshift sources with “V-shaped” rest-frame UV–optical spectral energy distributions and, very often, broad Balmer emission lines [Matthee et al., 2024, Labbé et al., 2025, Greene et al., 2024, Kocevski et al., 2025]. They are abundant: $n \simeq 10^{-5}\text{--}10^{-4}$ cMpc $^{-3}$ at $z \simeq 5$ [Matthee et al., 2024, Akins et al., 2025], orders of magnitude above the luminous-quasar density at the same epoch. Loiacono et al. [2026] show there is no significant evolution in the abundance of LRDs with $L_{\text{bol}} \gtrsim 3 \times 10^{44}$ erg s $^{-1}$ over $z > 2$, reaching $n = 3.4^{+5.6}_{-2.4} \times 10^{-5}$ Mpc $^{-3}$ at cosmic noon.

Three observational facts drive the whole debate. First, LRDs are extremely X-ray weak: Ananna et al. [2024] and Tortosa et al. [2026] find non-detections individually and in stacks, and Sacchi and Bogdán [2025] stacked 55 LRDs in the *Chandra* Deep Fields for nearly 400 Ms and still recovered nothing, which rules out a population of unobscured, intrinsically luminous super-Eddington accretors.

Second, they are radio silent: [Perger et al. \[2025\]](#) stacked 919 LRDs in VLASS and FIRST and found no emission above 3σ noise levels of ~ 11 and $\sim 18 \mu\text{Jy beam}^{-1}$ respectively, while [Akins et al. \[2025\]](#) stacked 434 LRDs in COSMOS-Web with MeerKAT (1.4 GHz) and the VLA (3 GHz) and derived a radio-loudness upper limit $R_{4400} = L_{5\text{ GHz}}/L_{4400\text{ Å}} \leq 10$ [in the sense of [Kellermann et al., 1989](#)]; [Mazzolari et al. \[2026\]](#) collect all the stacking attempts and note that the deepest reach $0.5\text{--}0.1 \mu\text{Jy}$ at $z \sim 5\text{--}6$. Third, the best spectra show enormous Balmer breaks together with Balmer *absorption* [[Naidu et al., 2025](#), [Rusakov et al., 2026](#)], which no ordinary stellar population produces.

Two 2026 results reframe the field. [Rusakov et al. \[2026\]](#), in *Nature*, argue that the line broadening is caused by electron scattering in a dense ionized cocoon of light-day scale rather than by Doppler motion, so that the intrinsic line cores imply $M_{\text{BH}} = 10^{5-7} M_{\odot}$ — two orders of magnitude below previous estimates — and the cocoon simultaneously explains the weak radio and X-ray signatures. [Barrufet et al. \[2026\]](#) find that the LRD fraction rises strongly with $\text{H}\alpha$ luminosity, which disfavors a simple orientation-based unification with “little blue dots” and instead supports gas-cocoon models in which column density drives the transition. Finally, [Rodríguez et al. \[2026\]](#) report the first radio-bright local LRD analogue, J2048 at $z = 0.4332$, with $S_{3\text{ GHz}} = 3.0 \pm 0.2 \text{ mJy}$, $\alpha = -0.39 \pm 0.04$ and $L_{\text{radio}} = 1.2 \times 10^{41} \text{ erg s}^{-1}$, and propose that inverse-Compton cooling by ultraviolet photons dims the synchrotron emission of the high-redshift population while leaving the mechanical output intact. That proposal is the pivot of Sect. 5.

3 The hypotheses currently being debated

Table 1 lists the nine distinct proposals that are alive in the literature as of August 2026, grouped by what the central object is.

Two remarks on the state of the debate. The evidence has moved decisively against H1 in its pure form: the dust reservoirs are too small [[Rusakov et al., 2026](#), [Naidu et al., 2025](#)] and the X-ray weakness is too severe [[Sacchi and Bogdán, 2025](#), [Tortosa et al., 2026](#)]. And H9 in its pure form is disfavoured because broad Balmer lines and Balmer absorption are not produced by ordinary H II regions, although [Roy et al. \[2026\]](#) show that ultra-compact galaxies reproduce a real subset of the properties, which is why most reviews now describe LRDs as a mixed or hybrid population [[Vaida and Farber, 2026](#)]. H3, H4, H5 and H6 are variations on a single physical picture — a super-Eddington engine buried in a dense, scattering-dominated envelope — and differ mainly in the mass of the engine and the geometry of the outflow.

4 Which of them can accelerate cosmic rays

A site accelerates cosmic rays if it has (i) a converging flow or a reconnecting field to energize particles, (ii) enough magnetic confinement to keep them there while they gain energy, and (iii) losses slow enough that the gain wins. Applying that filter to Table 1 isolates four sites, which are the four columns of Table 2.

- (i) **Jet / polar funnel.** Present in H1–H5. [Kuze et al. \[2026\]](#) build the only explicit particle-acceleration model of an LRD to date: relativistic jets and outflows launched from the black hole propagate through low-density polar funnels inside the envelope, and dissipate there. Their fiducial parameters are $M_{\text{BH}} = 10^{6.5} M_{\odot}$, envelope mass $10^{6.5} M_{\odot}$, dissipation radius $r_{\text{dis}} = 10^{16} \text{ cm}$, jet Lorentz factor $\Gamma_j = 2$, magnetic energy fraction $\epsilon_B = 0.01$, gyrofactor $\eta = 10$, a proton acceleration efficiency $\epsilon_p = 0.1$ and an ϵ_p^{-2} injection spectrum; they quote a comoving field $B' \simeq 1.3 \times 10^2 \text{ G}$ for a strongly collimated funnel.
- (ii) **Fast wind driving into the dense envelope.** This is the core of H6: “the engine may also launch fast winds that crash into the existing envelope to generate shocks” [[Naidu et al., 2026](#)]. It is observationally supported — [Korber et al. \[2026\]](#) report direct evidence for a fast AGN-driven outflow in a $z = 6.64$ LRD host with a kinetic luminosity $\sim 10^{43} \text{ erg s}^{-1}$, and

Rodríguez et al. [2026] argue for powerful mechanical outflows in the local analogue. H5 makes the same prediction by construction, since SS 433’s jets inflate the W50 nebula.

- (iii) **Accretion-disc corona.** Present in H1–H4, H7. Magnetic reconnection in a hot corona does accelerate particles, but as Table 2 shows the radiation energy density at $10 R_g$ exceeds the magnetic energy density by 3×10^5 , so the maximum electron energy collapses to ~ 10 GeV and nothing escapes. This site is a heater, not a cosmic-ray source.
- (iv) **Host-galaxy supernovae.** Present in H9 and in the hybrid readings of H3 and H6. This is the textbook cosmic-ray channel [Bell, 1978, Blandford and Ostriker, 1978, Drury, 1983]; the relevant point here is that Mazzolari et al. [2026] show the deepest SKAO observations will test exactly this scenario, probing $\text{SFR} \sim 5\text{--}50 M_\odot \text{yr}^{-1}$ at $z \sim 5\text{--}6$.

H7 and H8 are marked “marginal” because their defining feature is a *quasi-hydrostatic, convective, radiation-dominated* envelope [Pacucci et al., 2026, Gentile et al., 2026, Amaro Seoane, 2026]: there is no supersonic converging flow, so first-order Fermi acceleration has nothing to work with, even though the envelope is magnetized.

5 How to estimate the cosmic-ray electron spectrum

Write the steady-state electron spectrum inside the source as

$$N(\gamma) d\gamma = K \gamma^{-s} d\gamma, \quad \gamma_{\min} \leq \gamma \leq \gamma_{\max}. \quad (1)$$

Estimating it means fixing four things: the index s , the two ends $\gamma_{\min}, \gamma_{\max}$, and the normalization K . Everything below is verified symbolically in `verify_algebra.py`.

5.1 The index

There are two independent routes, and they should be used as a consistency check on each other.

Theory (forward). For diffusive shock acceleration the momentum distribution is $f(p) \propto p^{-q}$ with $q = 3r/(r-1)$, r the compression ratio [Bell, 1978, Blandford and Ostriker, 1978, Drury, 1983]. Since $N(E) dE \propto p^2 f(p) dp$, the energy index is

$$s = \frac{r+2}{r-1}, \quad r = \frac{(\gamma_{\text{ad}} + 1)\mathcal{M}^2}{(\gamma_{\text{ad}} - 1)\mathcal{M}^2 + 2} \implies s(\mathcal{M}) = \frac{2(\mathcal{M}^2 + 1)}{\mathcal{M}^2 - 1} \quad (\gamma_{\text{ad}} = 5/3), \quad (2)$$

so $s \rightarrow 2$ for a strong shock and $s = 2.17, 2.50, 3.33$ for $\mathcal{M} = 5, 3, 2$. The wind shock of site (ii) has $\mathcal{M} \simeq 198$ against 10^4 K gas (Table 2), i.e. $r = 4.000$ and $s = 2.000$ to four figures. This is why Kuze et al. [2026] adopt ε^{-2} for the protons; the same argument fixes the electrons. Mildly relativistic and oblique shocks steepen this to $s \simeq 2.2\text{--}2.4$, and supernova remnants empirically sit at $s \simeq 2.1\text{--}2.4$.

Observation (inverse). For an optically thin synchrotron source the flux index α (with $S_\nu \propto \nu^\alpha$) and s are locked:

$$\alpha = -\frac{s-1}{2} \iff s = 1 - 2\alpha. \quad (3)$$

Equation (3) is verified in `verify_algebra.py` by two independent routes (a δ -function kernel and the exact $F(\nu/\nu_c)$ kernel). *Caveat:* applied to the only LRD analogue with a measured radio slope, $\alpha = -0.39 \pm 0.04$ for J2048 [Rodríguez et al., 2026], this returns $s = 1.78$, which is flatter than any shock can produce — and the authors themselves attribute the slope to *optically thick* synchrotron, for which Eq. (3) does not hold. This is the single most important practical warning in this section: in an LRD, self-absorption and free-free absorption in the cocoon corrupt the spectral index, so s must come from theory and the radio slope can only be used once the source is demonstrably optically thin.

Cooling break. Wherever radiative losses beat escape, the steady-state solution of $\partial_\gamma(\dot{\gamma}N) = -Q$ with $\dot{\gamma} = -b\gamma^2$ and injection $Q \propto \gamma^{-s}$ is

$$N(\gamma) = \frac{Q_0}{b(s-1)} \gamma^{-(s+1)}, \quad b = \frac{4\sigma_T}{3m_e c} (U_B + U_{\text{rad}}), \quad (4)$$

i.e. the index steepens by exactly one above $\gamma_b = 1/(bt)$. In an LRD cocoon b is enormous and γ_b is of order unity, so the *observable* population is the cooled one, with $s+1 \simeq 3$.

5.2 The two ends of the spectrum

γ_{min} is set by injection out of the thermal pool and is of order a few; it matters only for $s > 2$, where it controls the energy budget. γ_{max} is the smaller of two limits, both verified symbolically:

$$\text{confinement (Hillas): } E_{\text{max}}[\text{eV}] = 299.79 \beta B[\text{G}] R[\text{cm}], \quad (5)$$

$$\text{radiative burn-off: } E_{\text{max}} = m_e c^2 \sqrt{\frac{3eB}{4\eta\sigma_T U_{\text{tot}}}}, \quad U_{\text{tot}} = U_B + U_{\text{rad}}, \quad (6)$$

where $\eta \geq 1$ is the gyrofactor (Bohm limit $\eta = 1$; Kuze et al. 2026 take $\eta = 10$). Equation (6) reduces, for $U_{\text{tot}} = U_B = B^2/8\pi$, to $E_{\text{max}} = m_e c^2 \sqrt{6\pi e/(\eta\sigma_T B)} \propto B^{-1/2}$, and the corresponding *synchrotron photon* energy is independent of B ,

$$E_\gamma^{\text{burn}} = \frac{27}{8} \frac{m_e c^2}{\alpha_f \eta} \simeq \frac{236 \text{ MeV}}{\eta}, \quad (7)$$

the familiar $\sim 10^2$ MeV synchrotron limit (the exact coefficient depends on the convention adopted for t_{acc} ; ours is $t_{\text{acc}} = \eta E/eBc$). In an LRD, $U_{\text{rad}} \gg U_B$, so both E_{max} and E_γ^{burn} fall by $\sqrt{U_B/U_{\text{tot}}}$. This is the quantitative form of the radio-dimming mechanism proposed by Rodríguez et al. [2026].

5.3 Normalization, route A: forward from the energy budget

Integrating Eq. (1) over energy,

$$U_e = \int_{\gamma_{\text{min}}}^{\gamma_{\text{max}}} K \gamma^{-s} \gamma m_e c^2 d\gamma = K m_e c^2 \frac{\gamma_{\text{max}}^{2-s} - \gamma_{\text{min}}^{2-s}}{2-s} \xrightarrow{s \rightarrow 2} K m_e c^2 \ln \frac{\gamma_{\text{max}}}{\gamma_{\text{min}}}, \quad (8)$$

the $s = 2$ limit being the one that actually applies. The recipe is then three steps:

1. *Get the mechanical power L_{mech} .* For the jet site, Kuze et al. [2026] set $L_j = L_{\text{Edd}} = 10^{44.6} \text{ erg s}^{-1}$ for $M_{\text{BH}} = 10^{6.5} M_\odot$. For the wind site, use the measured outflow kinetic luminosity: Korber et al. [2026] give $\sim 10^{43} \text{ erg s}^{-1}$. For the stellar site, use $L_{\text{mech}} \simeq 10^{42} (\text{SFR}/M_\odot \text{ yr}^{-1}) \text{ erg s}^{-1}$ from the supernova rate.
2. *Adopt an electron acceleration efficiency ϵ_e [A].* There is no LRD measurement of ϵ_e ; $\epsilon_e = 0.01$ – 0.1 brackets what is used for shocks elsewhere, and Kuze et al. [2026] adopt $\epsilon_p = 0.1$ for protons in this very system. Then $L_e = \epsilon_e L_{\text{mech}}$.
3. *Convert to K through $U_e V = L_e t$, with $t = \min(t_{\text{dyn}}, t_{\text{cool}})$ and V the emitting volume:*

$$K = \frac{L_e t}{V m_e c^2 \ln(\gamma_{\text{max}}/\gamma_{\text{min}})} \quad (s = 2). \quad (9)$$

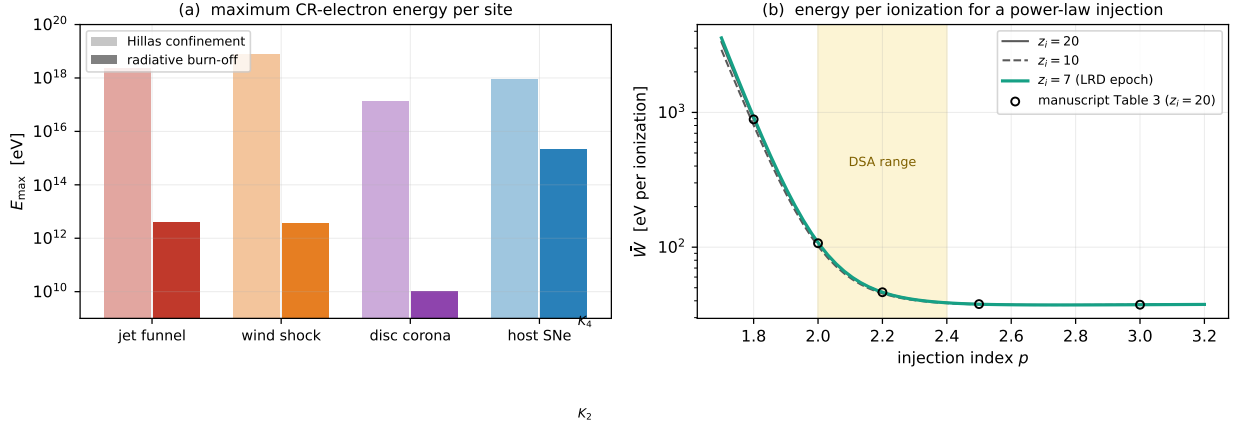


Figure 1: (a) Maximum cosmic-ray electron energy per site. Confinement (Hillas, Eq. 5) is never the binding constraint; radiative burn-off (Eq. 6) always is, and in the corona it collapses to 10 GeV. Dotted and dashed lines mark K_2 and K_4 , the regime boundaries of the companion manuscript. (b) Spectrum-averaged energy per ionization $\bar{W}(p)$ from the cascade yield table, at the LRD epoch ($z_i = 7$, green) and at the two redshifts tabulated in the manuscript. Open circles are the published Table 3 values, reproduced here to 0.14%. The shaded band is the DSA range $s = 2.0$ – 2.4 . Produced by `make_lrd_figures.py`.

5.4 Normalization, route B: inverse, from the radio non-detections

Converting a flux limit to a luminosity requires the K-correction

$$S_\nu = \frac{(1+z)^{1+\alpha} L_\nu(\nu_{\text{obs}})}{4\pi D_L^2}, \quad (10)$$

and the crucial step, which is where most of the physics sits, is that in fast cooling only a fraction $U_B/(U_B + U_{\text{rad}})$ of the electron power comes out as synchrotron radiation, the rest being inverse-Compton:

$$L_e = L_{\text{sync}} \left(1 + \frac{U_{\text{rad}}}{U_B} \right). \quad (11)$$

A synchrotron upper limit therefore translates into an electron-power upper limit that is *weaker* by $1 + U_{\text{rad}}/U_B$. If instead one has a detection and wants the minimum energy content, the classical minimum-energy condition follows from minimizing $U_e + U_B$ at fixed L_{sync} and gives, exactly,

$$\frac{U_B}{U_e} = \frac{s+1}{4} \quad \left(= \frac{3}{4} \text{ for } s = 2 \right). \quad (12)$$

5.5 Worked fiducial

Table 2 evaluates the four sites for an LRD at $z = 5$ with $M_{\text{BH}} = 10^{6.5} M_\odot$ ($L_{\text{Edd}} = 3.98 \times 10^{44} \text{ erg s}^{-1}$), $L_{\text{bol}} = 10^{45} \text{ erg s}^{-1}$ and an envelope of radius $R = 10^{16} \text{ cm}$ (3.9 light-days, matching the light-day scale of Rusakov et al. 2026). All entries come from `lrd_cr_numbers.py`.

Three results follow immediately.

The radio limits do not bite. At $z = 5$, $D_L = 4.77 \times 10^4 \text{ Mpc}$, so the $11 \mu\text{Jy}$ VLASS 3σ stack limit of Perger et al. [2025] corresponds to a rest-frame $L_{1.4 \text{ GHz}} \leq 1.0 \times 10^{25} \text{ W Hz}^{-1}$ (for $\alpha = -0.7$) and, integrated over 0.1–100 GHz, to $L_{\text{sync}} \leq 1.5 \times 10^{42} \text{ erg s}^{-1}$, i.e. 0.15% of L_{bol} . Applying Eq. (11) with the wind-site value $U_{\text{rad}}/U_B = 100$ gives $L_e \leq 1.6 \times 10^{44} \text{ erg s}^{-1}$, or 0.16 L_{bol} ; for the jet site the bound is 0.39 L_{bol} . The forward estimate of Sect. 5.3 with $\epsilon_e = 0.01$ gives $L_e = 10^{41} \text{ erg s}^{-1}$ (wind)

to $4 \times 10^{42} \text{ erg s}^{-1}$ (jet). *The observational bound is three orders of magnitude above the prediction.* The radio silence of LRDs is therefore not evidence against a cosmic-ray electron population; it is a statement about U_B/U_{rad} , exactly as [Rodríguez et al. \[2026\]](#) argue.

Neither do the X-ray limits, for a different reason. The power that does not emerge as synchrotron emerges as inverse Compton on the $T \sim 5000 \text{ K}$ photospheric field [[Gentile et al., 2026](#)], whose peak is at 1.2 eV. For a cooled $s = 2$ population the inverse-Compton continuum is flat in νF_ν and spans $\simeq 11.8$ decades from 6.5 eV to the Klein–Nishina turnover, so 5.9% of L_e falls in 2–10 keV. The fiducial wind case gives $L_{2-10} = 5.9 \times 10^{39} \text{ erg s}^{-1}$, a factor 53 below the stacked *Chandra* limit $\log L_{2-10} \lesssim 41.5$; an aggressive jet case ($\epsilon_e = 0.1$ of L_{Edd}) gives $2.3 \times 10^{42} \text{ erg s}^{-1}$, a factor 7.5 *above* it. [A] My reading is that this does not exclude the aggressive case, because the same Compton-thick column ($N_{\text{H}} = 10^{24}–10^{25} \text{ cm}^{-2}$) invoked to explain the X-ray weakness [[Tortosa et al., 2026](#), [Pacucci et al., 2026](#)] absorbs the 2–10 keV band regardless of its origin. The conclusion is that the cosmic-ray electron content of LRDs is at present observationally *unconstrained* from either side, and only the forward energetics and the IGM consequences (Sect. 6) can be used.

Escape. A Compton-thick envelope has a grammage of $1.7–17 \text{ g cm}^{-2}$; at the minimum-ionizing rate of $\sim 2 \text{ MeV (g cm}^{-2})^{-1}$ this stops electrons below roughly 30–300 MeV. Electrons above that traverse the envelope, so a $\gamma_{\text{max}} \sim 10^6$ population loses only its low-energy end on the way out. The escaping fraction $f_{\text{esc}}^{\text{CR}}$ into the IGM must still be treated as a free parameter [A], since it depends on the magnetic topology of the host, which no LRD observation constrains.

6 Closing the loop: what this delivers to the IGM

The companion manuscript computes, for an electron injected at z_i with kinetic energy K_{ini} , the total number of ionizations $Y(K_{\text{ini}}, z_i)$ summed over every generation of the cascade, and the spectrum-averaged energy per ionization $\bar{W}$ for a power-law injection. Evaluating that machinery on the $z_i = 7$ row — the lowest redshift at which the table is populated, and the LRD epoch — gives

$$\bar{W}(z_i = 7) = 925, 108, 46.3, 37.7, 37.4 \text{ eV} \quad \text{for } p = 1.8, 2.0, 2.2, 2.5, 3.0, \quad (13)$$

and the same code reproduces the published Table 3 values at $z_i = 20$ and $z_i = 10$ to a maximum deviation of 0.14% (Fig. 1b). For the DSA index $s = 2$ the relevant value is $\bar{W} = 108 \text{ eV}$. Note that $p = 1.8$ must *not* be used here: for $p < 2$ the integral is dominated by K_{max} , where the $z_i = 7$ yields are truncated by the $z = 5.5$ end of the integration.

The comoving energy per hydrogen atom deposited by the LRD population is

$$\epsilon_{\text{H}} = f_{\text{esc}}^{\text{CR}} \frac{n_{\text{LRD}} L_e t_{\text{era}}}{n_{\text{H}}}, \quad \frac{N_{\text{ion}}}{n_{\text{H}}} = \frac{\epsilon_{\text{H}}}{\bar{W}}, \quad \Delta T = \frac{N_{\text{ion}}}{n_{\text{H}}} \times (4.3–6.9) \times 10^4 \text{ K}, \quad (14)$$

the last relation being the heat–ionization lock derived in Sect. 9.4 of the manuscript, inverted. With $n_{\text{H}} = 1.944 \times 10^{-7} \text{ cm}^{-3}$ comoving and $t_{\text{era}} = 800 \text{ Myr}$ ($z = 7 \rightarrow 4$), Table 3 gives the result.

Three readings of Table 3. First, the realistic rows sit at $N_{\text{ion}}/n_{\text{H}} \sim 10^{-6}–10^{-4}$, one to three orders of magnitude below the $2.3 \times 10^{-4}–4.6 \times 10^{-3}$ that the manuscript obtains for the hadronic $\Delta T = 10–200 \text{ K}$ scale of [Leite et al. \[2017\]](#): LRDs are not a competitive ionizing source, and by the argument of Sect. 9.4 of the manuscript their net effect on the photon budget [[Muñoz et al., 2024](#)] is again to *widen* it, by a negligible amount. Second, the ΔT column is the interesting one, because $\Delta T \sim 1–18 \text{ K}$ is comparable to or above the adiabatic floor $T_{\text{ad}}(z = 7) = 1.33 \text{ K}$, which is the regime the 21-cm experiments are sensitive to: LOFAR disfavours mean gas temperatures below 44 K at $z = 9.1$ [[Mertens et al., 2025](#), [Ghara et al., 2025](#)] and HERA requires the gas above the adiabatic floor by $z = 10.4$ [[HERA Collaboration, 2023](#)]. Third, the last row — $\Delta T \sim 5 \times 10^3 \text{ K}$ — would be flatly inconsistent with the measured $T_0 \simeq 1.0–1.2 \times 10^4 \text{ K}$ at $z = 5.4–5.8$ [[Gaikwad et al., 2020](#)] for gas that is only then being photoheated, and is therefore excluded. [A] That exclusion is itself a

result: the thermal history caps the product $n_{\text{LRD}} \epsilon_e f_{\text{esc}}^{\text{CR}}$, and it does so far more tightly than any radio or X-ray stack.

7 Verification

`verify_algebra.py` checks fifteen results with SYMPY, each by symbolic manipulation rather than numerical spot-checking: the strong-shock compression ratio and Eq. (2); Eq. (3) by two independent kernels; the cooling steepening of Eq. (4); the normalization integral of Eq. (8) including the $s \rightarrow 2$ limit; the de-boosting factor of Eq. (11); the K-correction exponent of Eq. (10); the minimum-energy ratio of Eq. (12); the two E_{max} expressions and the B -independence of Eq. (7); and the internal consistency of Eq. (14) with the manuscript’s lock. Unit conversions are checked numerically: the Hillas prefactor evaluates to $299.8 \text{ eV G}^{-1} \text{ cm}^{-1}$ and Eq. (7) to 236.3 MeV . `lrd_cr_numbers.py` reproduces Table 3 of the manuscript to 0.14% as an end-to-end regression on the figure and table values generated previously.

8 Summary

1. Nine hypotheses for LRDs are alive as of August 2026 (Table 1); the field has converged on a super-Eddington engine inside a dense, scattering-dominated envelope, with the mass of the engine (10^3 – $10^7 M_{\odot}$) and the geometry of the outflow the open questions.
2. Four of them contain a genuine cosmic-ray acceleration site: the polar jet funnel, the fast wind driving into the envelope, and (for the stellar readings) host supernovae. The disc corona is a heater, not an accelerator; the quasi-star and supermassive-star models have no supersonic converging flow at all.
3. The index follows from shock physics, $s = (r + 2)/(r - 1) \rightarrow 2$, steepening to $s + 1 \simeq 3$ above the cooling break; it must *not* be read off the radio slope, which is optically thick in the one case where it has been measured. The normalization follows either from $L_e = \epsilon_e L_{\text{mech}}$ or, as an upper bound, from the radio stacks corrected by $1 + U_{\text{rad}}/U_B$.
4. That correction factor is 10^2 – 10^5 , so the radio non-detections bound the electron power three orders of magnitude above the fiducial prediction. The X-ray stacks come within a factor of a few of the aggressive case but are evaded by the Compton-thick column. The tightest real constraint on the cosmic-ray electron output of LRDs is the measured IGM thermal history, through Eq. (14).

9 Publications cited, with online access

Reference	Cited for	Free access
Discovery and demographics		
Matthee et al. [2024]	LRD definition; $n \sim 10^{-5}$ – $10^{-4} \text{ cMpc}^{-3}$	https://arxiv.org/abs/2306.05448
Labbé et al. [2025]	candidate red AGN at $3 < z < 7$	https://arxiv.org/abs/2306.07320
Greene et al. [2024]	spectroscopic confirmation of broad lines	https://arxiv.org/abs/2309.05714
Kocevski et al. [2025]	LRD sample in the JWST legacy fields	https://arxiv.org/abs/2404.03576
Akins et al. [2025]	COSMOS-Web overabundance; $R_{4400} \leq 10$	https://arxiv.org/abs/2406.10341

Reference	Cited for	Free access
Loiacono et al. [2026]	no number-density evolution at $z > 2$	https://arxiv.org/abs/2606.30253
Vaida and Farber [2026]	2026 mini-review of the competing scenarios	https://arxiv.org/abs/2601.00089
The nine hypotheses		
Naidu et al. [2025]	black-hole star (H2)	https://arxiv.org/abs/2503.16596
Rusakov et al. [2026]	dense ionized cocoon, $M_{\text{BH}} = 10^{5-7} M_{\odot}$ (H3)	https://arxiv.org/abs/2503.16595
Barrufet et al. [2026]	LRD/LBD regimes support cocoon models (H3)	https://arxiv.org/abs/2608.21522
Liu et al. [2025]	Balmer break from super-Eddington accretion (H4)	https://arxiv.org/abs/2507.07190
Liu et al. [2026]	optically thick supercritical outflow (H4)	https://arxiv.org/abs/2607.17448
Zhou et al. [2026]	supermassive SS 433 analogue (H5)	https://arxiv.org/abs/2606.21105
Naidu et al. [2026]	η Car / Type IIn engine; wind shocks (H6)	https://arxiv.org/abs/2606.30711
Pacucci et al. [2026]	direct-collapse black hole (H7)	https://arxiv.org/abs/2601.14368
Gentile et al. [2026]	quasi-star model; $T \sim 5000$ K photosphere (H8)	https://arxiv.org/abs/2606.06575
Amaro Seoane [2026]	nested supermassive star (H8)	https://arxiv.org/abs/2607.20617
Shi et al. [2026]	black-hole stars from star–BH collisions (H8)	https://arxiv.org/abs/2608.27596
Roy et al. [2026]	ultra-compact star-forming galaxies (H9)	https://arxiv.org/abs/2606.23683
X-ray and radio constraints		
Ananna et al. [2024]	first X-ray view of LRDs	https://arxiv.org/abs/2404.19010
Sacchi and Bogdán [2025]	400 Ms <i>Chandra</i> stack; non-detection	https://arxiv.org/abs/2505.09669
Tortosa et al. [2026]	X-ray weakness vs. local AGN; N_{H}	https://arxiv.org/abs/2603.10162
Perger et al. [2025]	919 LRDs; 3σ of 11 and 18 $\mu\text{Jy beam}^{-1}$	https://arxiv.org/abs/2411.19518
Mazzolari et al. [2026]	compilation of all radio stacks; SKAO forecast	https://arxiv.org/abs/2606.30757
Rodríguez et al. [2026]	radio-bright local analogue; IC radio dimming	https://arxiv.org/abs/2608.16200
Cosmic-ray physics and outflows		
Kuze et al. [2026]	only explicit CR model of an LRD; jet funnel	https://arxiv.org/abs/2601.11203
Korber et al. [2026]	fast AGN-driven outflow, $L_{\text{kin}} \sim 10^{43} \text{ erg s}^{-1}$	https://arxiv.org/abs/2604.08687
Bell [1978]	diffusive shock acceleration	https://ui.adsabs.harvard.edu/abs/1978MNRAS.182..147B
Blandford and Ostriker [1978]	shock acceleration spectral index	https://ui.adsabs.harvard.edu/abs/1978ApJ...221L..29B
Drury [1983]	DSA review; $s = (r + 2)/(r - 1)$	https://ui.adsabs.harvard.edu/abs/1983RPPh...46..973D

Reference	Cited for	Free access
Hillas [1984]	confinement limit $E_{\max} = eBR\beta$	https://ui.adsabs.harvard.edu/abs/1984ARA%26A..22..425H
Rybicki and Lightman [1979]	synchrotron emissivity and $\alpha = -(s-1)/2$	https://ui.adsabs.harvard.edu/abs/1979rpa...book....R
Kellermann et al. [1989]	radio-loudness parameter R_{4400}	https://ui.adsabs.harvard.edu/abs/1989AJ....98.1195K
Novak et al. [2017]	SFR–radio luminosity calibration	https://arxiv.org/abs/1703.09724
Delvecchio et al. [2021]	$q_{\text{IR}}(M_*, z)$ infrared–radio correlation	https://arxiv.org/abs/2101.06192
Popesso et al. [2023]	star-forming main sequence	https://arxiv.org/abs/2203.10487
IGM and reionization context		
Planck Collaboration [2020]	cosmology; $\sigma(\tau) = 0.007$	https://arxiv.org/abs/1807.06209
Leite et al. [2017]	$\Delta T = 10\text{--}200$ K hadronic CR heating scale	https://arxiv.org/abs/1703.09337
HERA Collaboration [2023]	IGM above the adiabatic floor by $z = 10.4$	https://arxiv.org/abs/2210.04912
Mertens et al. [2025]	LOFAR power-spectrum upper limits	https://arxiv.org/abs/2503.05576
Ghara et al. [2025]	$\bar{T} < 44$ K disfavoured at $z = 9.1$	https://arxiv.org/abs/2505.00373
Gaikwad et al. [2020]	$T_0 \simeq 1.0\text{--}1.2 \times 10^4$ K at $z = 5.4\text{--}5.8$	https://arxiv.org/abs/2001.10018
Muñoz et al. [2024]	the photon budget crisis	https://arxiv.org/abs/2404.07250

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Table 1: Competing hypotheses for the nature of Little Red Dots. “Engine” is the object that supplies the luminosity. The two right-hand columns anticipate Sect. 4.

Hypothesis	Engine and key claim	Refs.
H1 Obscured / dust-reddened AGN	Standard SMBH with a dusty screen; the original reading	Matthee et al. 2024 , Labbé et al. 2025 , Kocevski et al. 2025
H2 Black-hole star (BH [∗])	SMBH inside a dust-free, dense, turbulent “atmosphere” that makes the Balmer break and absorption	Naidu et al. 2025
H3 Dense ionized cocoon around a low-mass SMBH	Electron scattering, not Doppler motion, broadens the lines; $M_{\text{BH}} = 10^{5-7} M_{\odot}$, near-Eddington	Rusakov et al. 2026 , Barrufet et al. 2026
H4 Super-Eddington / supercritical accretion	Puffed-up, self-shielding disc; intrinsic X-ray weakness and the Balmer break come from the accretion flow itself	Liu et al. 2025 , 2026
H5 Supermassive analogue of SS 433	Hyper-Eddington microquasar physics scaled up, viewed at high inclination; LBDs are the low-inclination counterparts	Zhou et al. 2026
H6 Intermediate-mass super-Eddington engine (η Car / Type IIIn analogue)	Slow dense wind forms a pseudo-photosphere; fast winds crash into it and generate shocks; $M < 10^5 M_{\odot}$	Naidu et al. 2026
H7 Direct-collapse black hole (DCBH)	Compressionally heated, collisionally ionized accretion flow onto a heavy seed in a pristine atomic-cooling halo	Pacucci et al. 2026
H8 Quasi-star / supermassive star	An SMBH seed inside a convective envelope radiating a $T \sim 5000$ K black body; or a hierarchically nested supermassive star trapping a nuclear star cluster	Gentile et al. 2026 , Amaro Seoane 2026 , Shi et al. 2026
H9 Compact star-forming galaxy (no AGN required)	Ultra-compact stellar cores, $R_{\text{eff}} < 300$ pc, $V_{\text{circ}} > 500$ km s ^{−1} ; reproduces the break and the UV slope but not the reddest continua or the broadest lines	Roy et al. 2026 , Vaida and Farber 2026

Table 2: The four candidate acceleration sites, evaluated for the fiducial LRD. $f_{\text{sync}} = U_B / (U_B + U_{\text{rad}})$ is the fraction of the electron power that emerges as synchrotron radiation. E_{max} from Eqs. (5) and (6) with $\eta = 1$.

site	B [G]	U_B [erg cm ^{−3}]	U_{rad}/U_B	s	$E_{\text{max}}^{\text{Hillas}}$ [eV]	$E_{\text{max}}^{\text{cool}}$ [eV]	f_{sync}
jet funnel	0.82	2.6×10^{-2}	251	2.0	2.4×10^{18}	4.2×10^{12}	4.0×10^{-3}
wind shock	2.6	2.7×10^{-1}	100	2.000	7.7×10^{16}	3.7×10^{12}	9.9×10^{-3}
disc corona	100	4.0×10^2	3.1×10^5	—	4.2×10^{16}	1.1×10^{10}	3.3×10^{-6}
host SNe	10^{-4}	4.0×10^{-10}	6.5	2.2	9.3×10^{17}	2.2×10^{15}	0.13

Table 3: Cosmic-ray electron deposition in the IGM from the LRD population, Eq. (14), with $\bar{W} = 108$ eV. The last row is an aggressive upper case ($\epsilon_e = 0.1$ of the Eddington jet power, full escape, high abundance) and is excluded by the measured thermal history.

n_{LRD} [cMpc $^{-3}$]	L_e [erg s $^{-1}$]	$f_{\text{esc}}^{\text{CR}}$	ε_{H} [eV]	$N_{\text{ion}}/n_{\text{H}}$	ΔT [K]
10^{-5}	10^{41}	0.1	2.8×10^{-4}	2.6×10^{-6}	0.11–0.18
10^{-5}	10^{41}	1	2.8×10^{-3}	2.6×10^{-5}	1.1–1.8
10^{-4}	10^{41}	1	2.8×10^{-2}	2.6×10^{-4}	11–18
10^{-5}	4.0×10^{43}	0.1	0.11	1.0×10^{-3}	44–70
10^{-4}	4.0×10^{43}	1	11.0	0.10	4400–7000